Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112145
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 26 Question Id : 691121551 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A new unit (α) of length is chosen such that it is equal to the speed of light in vacuum. What is the distance between Venus and Earth in terms of α units if light takes 6 min. 40 s to cover this distance ?

Options :

6911211871. $200\ \alpha$

6911211872. $400\ \alpha$

6911211873. $300\ \alpha$

6911211874. $500\ \alpha$

Question Number : 27 Question Id : 691121552 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the equation $H = \frac{x^p \epsilon^q E^r}{t^s}$

Where H = magnetic field; E = electric field, ϵ = permittivity, x = distance, t = time

The values of p, q, r and s respectively are :

Options :

6911211875. $1, 1, 1, 1$

6911211876. $-1, 1, 2, 1$

6911211877. $1, -1, -2, 1$

6911211878. $-1, -2, -2, 1$

Question Number : 28 Question Id : 691121553 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A car moving with a speed of 54 km/h takes a turn of radius 20 m. A simple pendulum is suspended from the ceiling of the car. Determine the angle made by the string of the pendulum with the vertical during the turning. (Take $g = 10\ m/s^2$)

Options :

6911211879. $\tan^{-1}(0.5)$

6911211880. $\tan^{-1}(0.75)$

6911211881. $\tan^{-1}(1.125)$

6911211882. $\tan^{-1}(0.25)$

Question Number : 29 Question Id : 691121554 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A gas balloon is going up with a constant velocity of 10 m/s. When this balloon reached a height of 75 m, a stone is dropped from it and balloon keeps moving up with the same velocity. The height of the balloon when the stone hits the ground is _____ m. (Take $g = 10\ m/s^2$)

Options :

6911211883. 85

6911211884. 150

6911211885. 129

6911211886. 125

Question Number : 30 Question Id : 691121555 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A thin biconvex lens is prepared from the glass ($\mu = 1.5$) both curved surfaces of which have equal radii of 20 cm each. Left side surface of the lens is silvered from outside to make it reflecting. To have the position of image and object at the same place, the object should be placed, from the lens at a distance of _____ cm.

Options :

6911211887. 10

6911211888. 12.5

6911211889. 13

6911211890. 13.5

Question Number : 31 Question Id : 691121556 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two identical bodies, projected with the same speed at two different angles cover the same horizontal range R. If the time of flight of these bodies are 5 s and 10 s, respectively, then the value of R is _____ m. (Take $g = 10\ m/s^2$)

Options :

6911211891. 250

6911211892. 25

6911211893. 500

6911211894. 125

Question Number : 32 Question Id : 691121557 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A solid cylinder having radius R and length L is slipping on a rough horizontal plane. At time $t = 0$ the cylinder has a translational velocity $v_0 = 49\ m/s$, perpendicular to its axis and a rotational velocity $v_0/4R$ about the centre. The time taken by the cylinder to start rolling is _____seconds. (coefficient of kinetic friction $\mu_K = 0.25$ and $g = 9.8\ m/s^2$)

Options :

6911211895. 15

6911211896. 5

6911211897. 10

6911211898. 7.5

Question Number : 33 Question Id : 691121558 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A liquid of density 600 kg/m^3 flowing steadily in a tube of varying cross-section. The cross-section at a point A is 1.0 cm^2 and that at B is 20 mm^2. Both the points A and B are in same horizontal plane, the speed of the liquid at A is 10 cm/s. The difference in pressures at A and B points is _____ Pa.

Options :

6911211899. 18

6911211900. 144

6911211901. 36

6911211902. 72

Question Number : 34 Question Id : 691121559 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A spherical liquid drop of radius R acquires the terminal velocity v_1 when falls through a gas of viscosity η. Now the drop is broken into 64 identical droplets and each droplet acquires terminal velocity v_2 falling through the same gas. The ratio of terminal velocities v_1/v_2 is _____.

Options :

6911211903. 4

6911211904. 0.25

6911211905. 32

6911211906. 16

Question Number : 35 Question Id : 691121560 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

One mole of diatomic gas having rotational modes only is kept in a cylinder with a piston system. The cross-section area of the cylinder is 4 cm^2. The gas is heated slowly to raise the temperature by 1.2 °C during which the piston moves by 25 mm. The amount of heat supplied to the gas is _____ J. (Atmospheric pressure = 100 kPa, $R = 8.3$ J/mol. K) (Neglect mass of the piston)

Options :

6911211907. 24.8

6911211908. 25

6911211909. 15.04

6911211910. 29.98

Question Number : 36 Question Id : 691121561 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Initial pressure and volume of a monoatomic ideal gas are P and V. The change in internal energy of this gas in adiabatic expansion to volume $V_{final} = 27\ V$ is _____ J.

Options :

6911211911. $-2\,PV\left(3\sqrt{3}-1\right)$

6911211912. $\frac{4}{3}PV$

6911211913. $-\frac{4}{3}PV$

6911211914. $\frac{3}{4}PV$

Question Number : 37 Question Id : 691121562 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The frequency of oscillation of a mass m suspended by a spring is v_1. If the length of the spring is cut to half, the same mass oscillates with frequency v_2. The value of v_2/v_1 is _____.

Options :

6911211915. 1

6911211916. 2

6911211917. $\sqrt{2}$

6911211918. $\sqrt{3}$

Question Number : 38 Question Id : 691121563 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A monochromatic source of light operating at 15 kW emits 2.5×10^{22} photons/s. The region of an electromagnetic spectrum to which the emitted electromagnetic radiation belongs to _____. (Take $h=6.6\times10^{-34}$ J.s and $c=3\times10^{8}$ m/s).

Options :

6911211919. Microwave

6911211920. Infrared

6911211921. Visible

6911211922. Ultraviolet

Question Number : 39 Question Id : 691121564 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A current carrying circular loop of radius 2 cm with unit normal $\hat{n}=\frac{\hat{k}+\hat{i}}{\sqrt{2}}$ is placed in a magnetic field, $\vec{B}=B_0\left(3\hat{i}+2\hat{k}\right)$. If $B_0=4\times10^{-3}$ T and current $I=100\sqrt{2}$ A, the torque experienced by the loop is _____ Wb.A. ($\pi=3.14$)

Options :

6911211923. $16\times10^{-5}\,\hat{k}$

6911211924. $5024 \times 10^{-7}\ \hat{k}$

6911211925. $5024 \times 10^{-7}\ \hat{i}$

6911211926. $5024 \times 10^{-7}\ \hat{j}$

Question Number : 40 Question Id : 691121565 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A 30 cm long solenoid has 10 turns per cm and area of 5 cm^2. The current through the solenoid coil varies from 2 A to 4 A in 3.14 s. The e.m.f. induced in the coil is $\alpha \times 10^{-5}$ V. The value α is _____.

Options :

6911211927. 60

6911211928. 12

6911211929. 120

6911211930. 34

Question Number : 41 Question Id : 691121566 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two point charges $q_1 = 3\ \mu C$ and $q_2 = -4\ \mu C$ are placed at points $\left(2\hat{i} + 3\hat{j} + 3\hat{k}\right)$ and $\left(\hat{i} + \hat{j} + \hat{k}\right)$ respectively. Force on charge q_2 is _____ N. $\left(\text{Take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ SI Units}\right)$

Options :

6911211931. $\left(12\hat{i} + 24\hat{j} + 24\hat{k}\right) \times 10^{-3}$

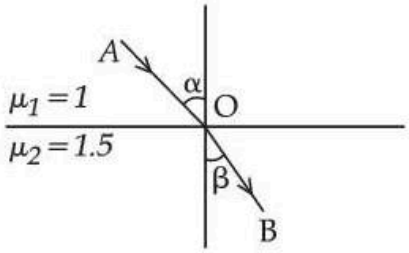
6911211932. $\left(4\hat{i} + 8\hat{j} + 8\hat{k}\right) \times 10^{-3}$

6911211933. $\left(3\hat{i} + 6\hat{j} + 6\hat{k}\right) \times 10^{-3}$

6911211934. $\left(-4\hat{i} - 8\hat{j} - 8\hat{k}\right) \times 10^{-3}$

Question Number : 42 Question Id : 691121567 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Light ray incident along a vector $\vec{AO}\left(\vec{AO} = 2\hat{i} - 3\hat{j}\right)$ emerges out along vector $\vec{OB}\left(\vec{OB} = C\hat{i} - 4\hat{j}\right)$ as shown in the figure below. The value of C is _____.



Options :

6911211935. 1.6

6911211936. 0.16

6911211937. 11.6

6911211938. 16

Question Number : 43 Question Id : 691121568 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

K_1 and K_2 be the maximum kinetic energies of photoelectrons emitted from a surface of a given material for the light of wavelength λ_1 and λ_2, respectively. If $\lambda_1 = 2\lambda_2$ then the work function of material is given by :

Options :

6911211939. $K_2 + 2K_1$

6911211940. $2K_2 - K_1$

6911211941. $K_1 - 2K_2$

6911211942. $K_2 - 2K_1$

Question Number : 44 Question Id : 691121569 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two radioactive substances A and B of mass numbers 200 and 212 respectively, shows spontaneous α-decay with same Q value of 1 MeV. The ratio of energies of α-rays produced by A and B is _____.

Options :

6911211943. $\frac{2548}{2650}$

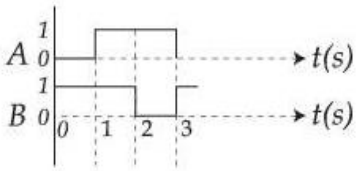
6911211944. $\frac{2706}{2646}$

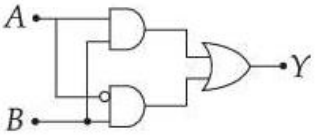
6911211945. $\frac{2597}{2600}$

6911211946. $\frac{2862}{2499}$

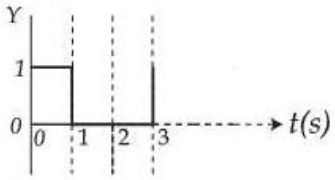
Question Number : 45 Question Id : 691121570 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

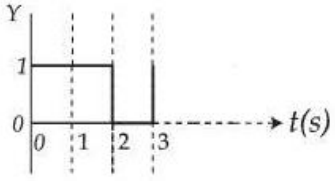
The output Y for the given inputs A and B to the circuit is :

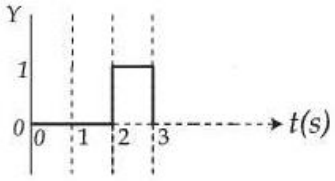


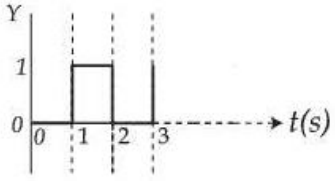


Options :

6911211947. 

6911211948. 

6911211949. 

6911211950. 

Physics Section B

Section Id :	69112146
Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory

Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112146
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 46 Question Id : 691121571 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A parallel plate capacitor is having separation between plates 0.885 mm. It has a capacitance of 1 μF when the space between the plates is filled with an insulating material of resistivity 1×10^{13} Ωm and resistance 17.7×10^{14} Ω. Relative permittivity of the insulating material is $\alpha\times10^{7}$. The value of α is _____.

(Take permittivity of free space $=8.85\times10^{-12}$ F/m)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 47 Question Id : 691121572 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Some distant star is to be observed by some telescope of diameter of objective lens a, at an angular resolution of 3.0×10^{-7} radian. If the wavelength of light from the star reaching the telescope is 500 nm, the minimum diameter of the objective lens of the telescope is _____ cm. (nearest interger)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

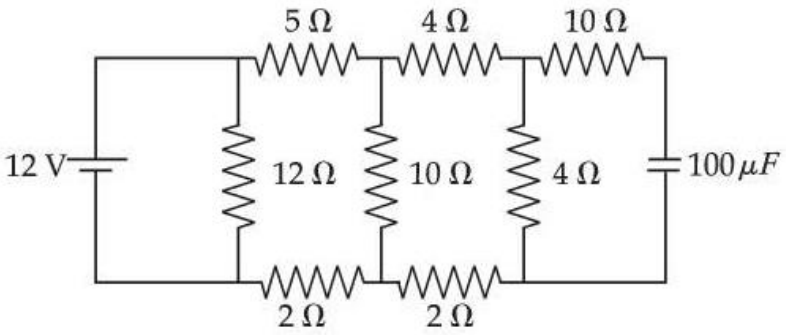
Question Number : 48 Question Id : 691121573 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A 5 mg particle carrying a charge of $5\pi\times10^{-6}\,C$ is moving with velocity of $\left(3\hat{i}+2\hat{k}\right)\times10^{-2}$ m/s in a region having magnetic field $\vec{B}=0.1\,\hat{k}$ Wb/m^2. It moves a distance of α meter along $\hat{k}$ when it completes 5 revolutions. The value of α is _____ .

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 49 Question Id : 691121574 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The stored charge in the capacitor in steady state of the following circuit is _____ μC.



Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 50 Question Id : 691121575 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Two masses of 3.4 kg and 2.5 kg are accelerated from an initial speed of 5 m/s and 12 m/s, respectively. The distances traversed by the masses in the 5th second are 104 m and 129 m, respectively. The ratio of their momenta after 10 s is $\frac{x}{8}$. The value of x is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Chemistry Section A

Section Id :	69112147
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112147
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 51 Question Id : 691121576 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical